Consistency by Construction: A Deterministic Policy-Guard Harness for a Low-Cost Agent on CAR-bench (Track 1)

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Abstract

CAR-bench scores agents with Pass^3: a task counts only if all three independent trials succeed, so inconsistency is penalized as heavily as incapability. Instead of buying consistency with a larger model, we wrap a low-cost model (`deepseek-v4-flash`) in a 3.3k-line deterministic policy guard that validates every draft response — tool-call schemas, confirmation state machines, physical and policy preconditions, evidence requirements for parameter values, and narrative honesty — before anything leaves the agent. Violations trigger at most two in-turn regenerations with targeted, error-time coaching; if the budget is exhausted, deterministic fallbacks strip illegal calls or produce an honest, topic-specific refusal, never a fabricated action. On the 125-task public test split our submission reaches 52.7% Pass^3 (est. 69.9% after excluding externally contaminated tasks), placing a sub-frontier model among frontier bare-agent baselines at under $4 per five-leg evaluation. We report two transferable findings: coaching *placement* matters more than content (always-on prompt coaching measurably hurts; error-time coaching helps), and guards themselves can over-trigger (tasks with $\geq$10 guard interventions pass at 34.8% vs. 79.4% below that threshold).

1 Introduction

CAR-bench [Kirmayr *et al.*, 2026] evaluates LLM agents as in-car voice assistants: 254 public tasks over 58 interconnected tools, 19 written domain policies, an LLM-simulated user, and three task families — *Base* (correct tool use and state changes), *Hallucination* (admitting missing capabilities instead of fabricating), and *Disambiguation* (resolving ambiguity via preferences or clarification before acting). The challenge metric is Pass^3 [Yao *et al.*, 2025]: each task is run three times and counts only if *all three* trials succeed, averaged over the three splits.

The benchmark's central observation is that current models favor *completion over compliance*: they act even when policy requires a precondition, a confirmation, a clarifying question, or a refusal. Its second observation is instability: organizer baselines lose 9–16 points going from Pass@1 to Pass^3. Both failure modes are only weakly responsive to model scale — the strongest baseline (Claude Opus 4.6) still scores 0.48 on Hallucination.

Our submission attacks both failure modes architecturally. We keep the agent shell thin (516 lines: A2A message handling and LiteLLM calls) and put the engineering weight into a *deterministic policy guard* (3,313 lines, 201 unit tests, 27 environment switches) that sits between the model and the evaluator and enforces, before any response is emitted, every deterministic check the evaluator will later score. The model proposes; the guard verifies; disagreements are resolved by bounded regeneration with error-specific coaching, and finally by honest refusal — never by the guard fabricating or repairing actions itself.

Contributions: (1) a policy-as-code guard architecture with a shadow state machine reconstructed purely from evaluator-visible traffic (§2); (2) an evaluation on the full public test split showing a low-cost model reaching frontier-baseline Pass^3 territory (§4); (3) pre-registered ablations, including a falsification result for always-on prompt coaching and a quantified "guard over-triggering" pathology, plus negative results we chose not to ship (§5).

2 System

2.1 Shadow state from visible traffic only

The evaluator is the sole executor of tools and owner of hidden state; the agent sees a system prompt, simulated-user turns, and tool results. A *shadow state machine* ingests exactly this visible traffic — parsing 40+ tool-result shapes (weather, windows, HVAC, lights, navigation, routes) plus the dated system prompt — into a best-effort model of the car. Every validator degrades gracefully: if the shadow state cannot support a check, the check is skipped rather than guessed. This keeps the guard within the organizer-sanctioned validator class (system prompt, trajectory, static tool catalog) with no probing of hidden evaluator state.

2.2 Rule families

Twelve deterministic rule families mirror the benchmark's written policies: **(1) whitelist** — tool names and arguments must match the task's declared schemas (unknown tools,

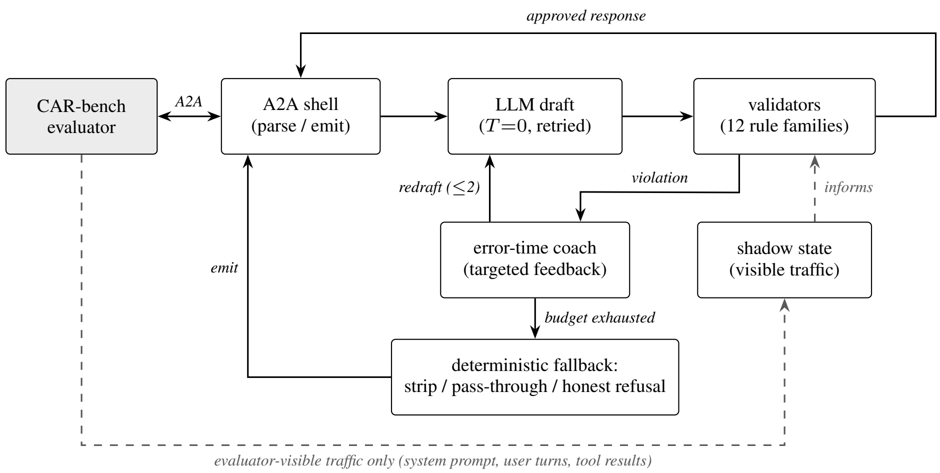

Figure 1: Architecture. The guard sits between the model and the A2A boundary: every draft passes 12 validator families backed by a shadow state built *only* from evaluator-visible traffic. Violations trigger at most two regenerations, each coached by targeted error feedback; if the budget is exhausted, deterministic fallbacks strip illegal calls, pass through legitimate clarification questions, or produce an honest refusal. The guard never fabricates or injects tool calls.

missing/extra/illegal parameters); **(2) confirmation** — irreversible actions (email, trunk, high beams) require a proposal turn with explicit parameters and an affirmative reply, judged by a deterministic affirmation classifier; **(3–8) preconditions** — weather checks before opening the sunroof, fog-light conditions, defrost/AC/fan/airflow interlocks, navigation-edit sequencing, same-day clamps on weather/calendar lookups, window-vs-AC energy confirmations; **(9) narrative honesty** — a two-layer check on text-only turns: a deterministic prescreen maps draft claims to a static catalog of tool-domain keywords, and only suspicious drafts go to a single $T=0$ LLM self-check that sees the *available* tools but is never told which tool is missing; **(10) preference-first evidence** — a parameter value must be traceable to an explicit user request, a retrieved preference, a written policy default, or observed state, checked against the actual `get_user_preferences` response text rather than a mere "preferences were consulted" flag; **(11) ask-gate** — the mirror rule on text: the agent may not ask the user for a concrete value (including menu-style multi-option asks) before exhausting preference and context evidence; **(12) no-retry** — a tool that returned a vendor "unavailable, do not retry" result is blocked for the rest of the session.

2.3 Bounded reprompts with error-time coaching

When validation fails and budget remains (default: two regenerations), the guard renders each violation as *targeted* feedback — what was blocked, why, and the specific corrective action — appended as a system message to a fresh copy of the conversation, and asks the model to redraft. Coaching text therefore appears *only at the moment of error*, a design choice validated by ablation (§5). Two auxiliary LLM roles exist: the narrative self-check above, and an optional self-consistency vote for disambiguation decisions (shipped off; §5).

2.4 Deterministic fallbacks, never fabrication

If the budget is exhausted, three fallbacks apply in order: *strip* illegal tool calls while keeping legal ones; *pass through* text flagged only by ask-gate rules (for disambiguation-style tasks, asking is often exactly the scored behavior); otherwise emit an *honest, topic-specific refusal* naming what cannot be done ("I can't help with the sunshade…") — across all five scored evaluation legs this refusal path fired 32 times and never coincided with a hallucination error. Two invariants hold everywhere: the guard never invents tool calls the model did not produce, and user-visible text must be consistent with the calls actually emitted. Determinism hardening (temperature 0.0 end-to-end with explicit `None`-safe resolution, retry-wrapped API calls, prompt caching) targets the residual trial-to-trial variance that Pass^3 punishes. All 27 guard switches are pinned in the submitted scenario TOML, making the evaluated configuration byte-reproducible.

3 Model Selection

We selected the agent LLM with a pre-registered bake-off on the 31-task disambiguation train split (single trial, identical guard): `deepseek-v4-flash` averaged 17.3/31 across three runs, against GLM-5.2 at 16, Kimi-K2.6 at 12, MiniMax-M2.5 at 10, and Qwen3.5-397B-A17B at 9. No challenger reached the pre-registered 20/31 switching threshold, so we kept the incumbent, with which the guard had been co-tuned. The result is consistent with our thesis: with a strong guard, provider choice within a price tier mattered less than guard quality.

4 Results

We evaluated the exact submitted configuration on the full public test split (125 tasks: 50 Base, 50 Hallucination, 25 Disambiguation) with three independent legs, synthesizing

System	Base	Hall.	Dis.	Avg. Pass^3
Claude Opus 4.6	.80	.48	.46	.58
GPT-5	.66	.60	.36	.54
Claude Sonnet 4	.63	.46	.32	.47
Gemini 2.5 Flash	.59	.41	.22	.41
Qwen3-32B	.45	.27	.22	.31
xLAM-2-32B	.26	.11	.12	.16
Ours (raw)	.68	.58	.32	**.527**
Ours (clean est.)	.68	.62	.80	**.699**

Table 1: Pass^3 by split. Top: organizer baselines (bare agents, public dev leaderboard). Bottom: our system on the 125-task public test split, three legs, `gemini-3.5-flash` simulator/judge; *raw* counts externally contaminated tasks as failures, *clean est.* excludes them (Disambiguation then rests on 8/10 tasks; see text). Judge configurations differ between blocks, so comparisons are indicative.

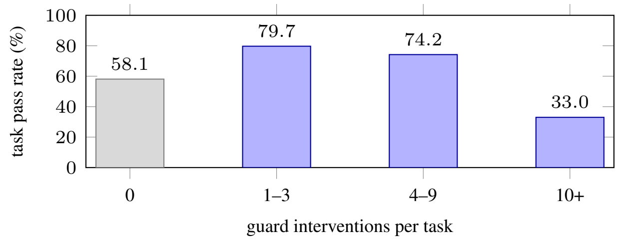

Figure 2: Guard interventions vs. pass rate (Base+Hallucination, 5 legs). Moderate guard activity coincides with the highest pass rates; heavy churn ($\geq$10) collapses them. The 0-intervention bucket is depressed by external API contamination (34.5% of its tasks), not by the guard.

Pass^3 as a cross-leg AND, under two simulator/judge variants: `gemini-2.5-flash` (as in the paper's setup) and `gemini-3.5-flash` (expected closer to the hidden evaluation). Total cost for all five scored legs was \$3.84 (\$2.56 agent-side) — the entire system runs on flash-tier pricing.

Table 1 shows the headline: 52.7% raw Pass^3 (Base .68, Hallucination .58, Disambiguation .32), with mean Pass@1 of 64.9%. During this run a shared, metered Gemini key ran dry mid-flight, which contaminated 16–96% of Disambiguation tasks (and 2–6% of Hallucination) across concurrent legs; excluding contaminated tasks yields an estimated 69.9%. We report both because neither is fully satisfying: the raw number charges real API failures against the agent, while the clean estimate rests on a reduced Disambiguation sample. The comparison we can make with confidence is Base+Hallucination, where results were stable across judges and legs. Against bare-agent baselines, a flash-tier model with a deterministic guard lands between GPT-5 and Claude Opus 4.6 raw, with Hallucination (.58) close to the best baseline observed (.60) — evidence that limit-awareness can be engineered rather than bought.

Judge sensitivity. In a controlled 12-task A/B (same code, same tasks), switching the simulator/judge from `gemini-2.5-flash` to `-3.5-flash` flipped exactly one task, traced to a genuine behavioral difference rather than judge noise; on a separate 15-task subset the two judges scored 12/15 vs. 15/15. Absolute numbers therefore shift by several points with judge choice, which should be kept in mind when comparing rows of Table 1.

5 Ablations and Lessons

Coaching placement: a pre-registered falsification. A bundle of three always-on coaching paragraphs (added to the system prompt) plus unified ask-gate guidance produced our first-ever wins on one hard task but dropped the disambiguation-train mean from 17.3/31 to 16.3 and Pass^3 from 13/31 to 12, destabilizing three previously reliable tasks. We pre-registered a decomposition: if the win vanished with the aggressive paragraphs off, the bundle hypothesis was falsified. It was — the rerun reproduced 16.3/12 with the hard task back at 0/3. The mechanism: always-on coaching that pushes the agent to *act* suppresses the legitimate "ask the user" exit that disambiguation tasks score. The shipped design therefore injects formatting/commitment coaching *only into the reprompt channel*, where it is conditioned on an actual error.

Guards can over-trigger. Across five legs the guard fired 3,350 times; two preference rules account for 70.5% (menu-style ask 1,410; value evidence 952). On contamination-clean Base+Hallucination tasks (n=500), tasks with $\geq$10 interventions passed at 34.8% vs. 79.4% below (Figure 2). Forensics traced the heavy-churn cluster to an evidence gate with no exhaustion exit: with genuinely empty user preferences, the agent could satisfy neither "use a preference" nor "proceed," churned until the simulated user ended the conversation, and failed with zero actions. The lesson we take: *every hard gate needs a deterministic exit* — we shipped the exemption for the one tool family where it was validated (empty-preference floor for airflow) and left the generalized version off pending per-family validation.

Things we built and did not ship. A self-consistency vote (3 samples at T=0.7) over the disambiguation meta-decision (*check preferences / check context / ask / proceed*) raised single-trial means from 17.3 to 19.3/31, but we shipped it disabled: per-turn cost rises, and we could not complete Pass^3-level validation of the gain before the deadline. Honest accounting of within-run variance (DeepSeek is not bit-exact even at T=0) is also why the reprompt budget is 2, not 1: one extra genuine self-correction opportunity measurably reduces "two bad samples in a row" failures without letting fallbacks guess values.

Engineering for Pass^3. Three infrastructure incidents (VM idle termination; a checkpoint-file race under 6-way concurrency, fixed by a one-line per-PID isolation patch after a data-integrity audit; and the API-quota exhaustion above, whose failure signature — reward 0.0 with empty reward info — is indistinguishable from genuine failure in the current tooling) cost more evaluation signal than any modeling choice. For consistency-graded benchmarks, quota headroom and contamination-aware harvesting are part of the method, not logistics.

6 Reproducibility and Artifact

Everything needed to rerun our submission is pinned. The agent image is public on GHCR and is referenced by digest (`sha256:8cb571f0...`, `linux/amd64`). It is built from the public repository by a GitHub Actions workflow. The scenario TOML lists all 27 guard switches, the temperature of 0.0, the reasoning-effort selector, and the model route. Nothing is left to an implicit default. The hidden-set container therefore boots into exactly the configuration we evaluated. The only secret is `DEEPSEEK_API_KEY`.

We verified these pinning claims directly. A startup audit in an empty environment, with only the TOML-declared variables injected, produced a healthy server and a live DeepSeek completion. The agent needs no proxy, no bundled `.env` file, and no credential from any other provider. As a last end-to-end check we ran the official Option-C validation flow. The organizer-published evaluator image drove the anonymously pulled public agent image on one task per split. All three tasks scored reward 1.0. The hallucination task ended with the guard's honest fallback ("I can't help with the sunshade right now…"), not with a fabricated action.

The whole loop is cheap to repeat. The five-leg public-test evaluation cost $3.84 in API spend, and $2.56 of that was agent-side. The full multi-week tuning campaign stayed under $10. Per-turn LLM latency in the final validation averaged about 15 seconds. A single machine and a hobbyist budget are enough to reproduce every experiment in this report.

7 Future Work

Three measurements come next. First, a guard-off ablation on the full public test split. We spent the pre-deadline budget on stabilizing the shipped configuration, so the guard's isolated contribution currently rests on baseline comparisons and intervention statistics. A controlled pair with `GUARD_ENABLED=false` will close that gap. Second, per-family validation of the generalized empty-preference floor, which we shipped default-off. The airflow family shows the exemption works where it has been validated. Fifteen other preference-gated families need the same A/B before the switch can default on. Third, an error analysis of the official hidden-set results once scores return. The organizers state that the hidden set removes gray areas. That analysis can then separate real capability gaps from the judge sensitivity measured in Section 4.

The design itself should travel. Policy-as-code validators, evidence gates, and coaching that appears only at the moment of error assume nothing specific to cars. Any benchmark that scores compliance all-or-nothing has the same structure. Porting the harness to related tool-agent suites is the natural next test of the thesis.

8 Compliance and Conclusion

The guard consumes only evaluator-visible traffic. It never probes hidden state, hardcodes answers, or modifies the evaluator. The official evaluator image is used unmodified. Source code is public after the deadline.[1]

CAR-bench argues that deployment-grade agents must be consistent, honest about limits, and policy-compliant. Our submission suggests these are substantially *systems* properties. A 3.3k-line deterministic guard moves a flash-tier model into the Pass^3 range of frontier bare agents at a small fraction of their cost. Its ingredients are policy-as-code validators, evidence gates, bounded coached regeneration, and honest fallbacks. The failure modes that remain are over-triggering gates without exits, judge sensitivity, and infrastructure fragility under concurrency. These are also systems problems. We expect further gains from treating them as such.

References

[Kirmayr *et al.*, 2026] Johannes Kirmayr, Lukas Stappen, and Elisabeth André. CAR-bench: Evaluating the consistency and limit-awareness of LLM agents under real-world uncertainty. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2026, San Diego, California, United States, July 2-7, 2026*, pages 40599–40618. Association for Computational Linguistics, 2026.

[Yao *et al.*, 2025] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik R. Narasimhan. τ-bench: A benchmark for Tool-Agent-User interaction in real-world domains. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025.

[1] https://github.com/WLeann/car-bench-ijcai-track1; agent `deepseek/deepseek-v4-flash` via LiteLLM (https://github.com/BerriAI/litellm), served at https://api.deepseek.com.